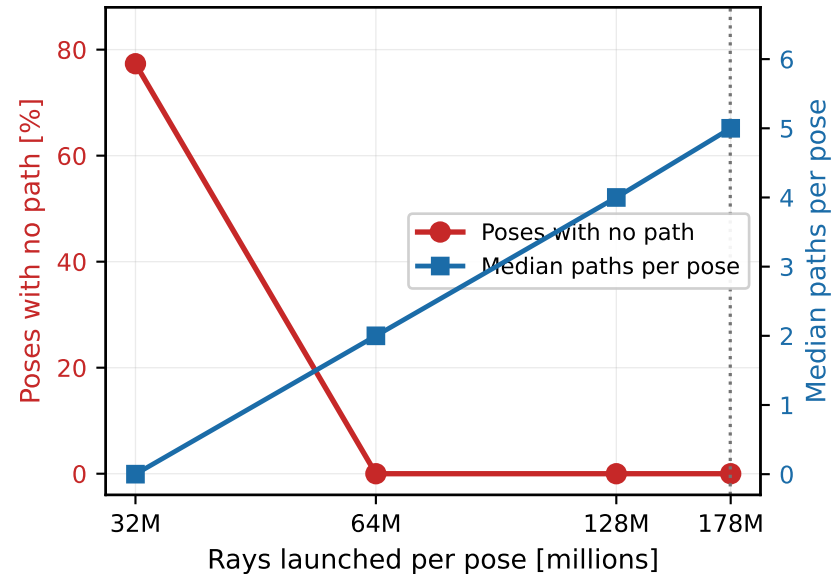
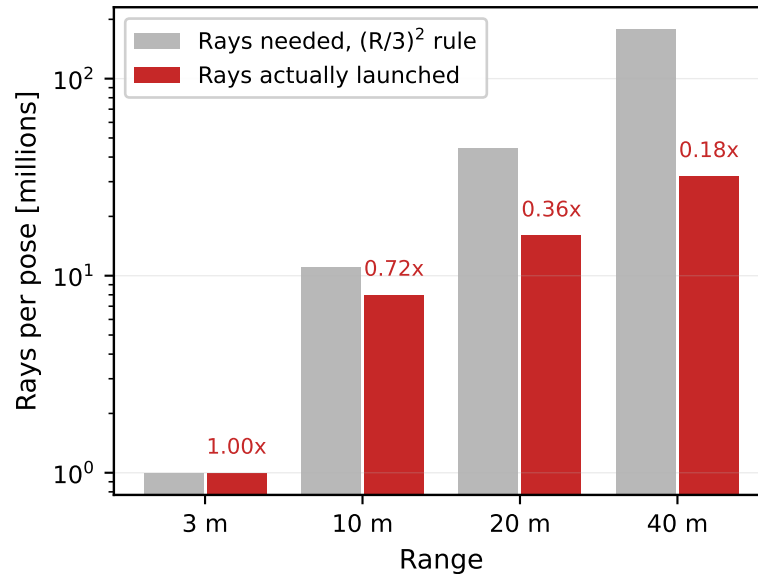


DJI Matrice 4E hovering, belly view, 3.5 GHz. Is range a wall for the path solver, or a budget?

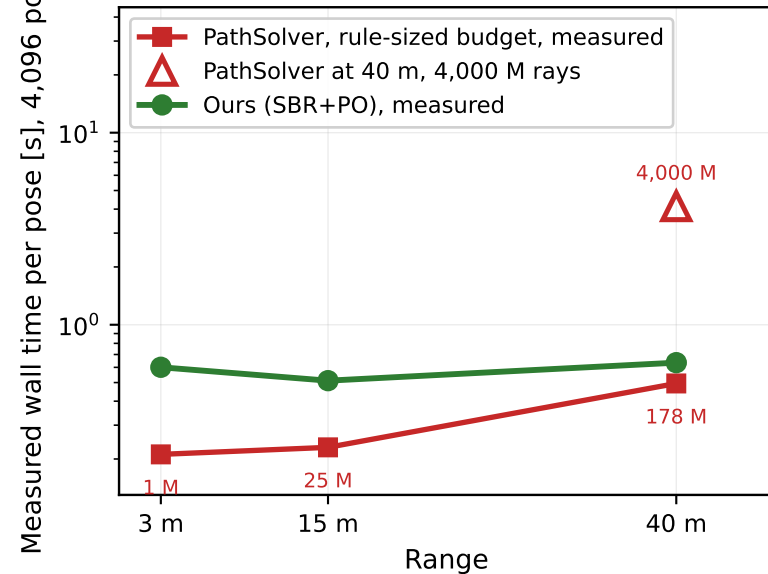
(a) PathSolver at 40 m, rays alone fix it
77% empty at 32M, 0% at 64M



(b) PathSolver budget: this sweep under-launched,
and the gap widens with range



(c) Measured cost, 4,096 poses on both arms.
Budget moves it 8.4x, range 2.3x



Panels (a) and (b) carry the path-solver arm alone: the two ledgers behind them hold only path-solver keys, so the SBR+PO arm has no range point there. Its cost comes from its own ledger and appears in panel (c). (a) Range, geometry, poses and rotor speeds are all held fixed at 40 m. Only the ray count changes, and the empty poses vanish, so the collapse at that range is a budget shortfall rather than a property of the solver. (b) The grey bars are the rule, rays scaled as $(R/3)^2$ from the 3 m case; the red bars are what this sweep launched. Its wall time is flat across range (0.167 s per pose at 3 m down to 0.147 s at 40 m, 512 poses) because the budget stayed near the 3 m value. (c) Measurement only, at rule-sized budgets; the rule itself is panel (b). Both arms run the same 4,096 poses, same airframe, azimuth 0 and elevation -15 deg, 3.5 GHz, 19.7 kHz slow time. PathSolver budgets follow the panel (b) rule (1 M at 3 m, 25 M at 15 m, 178 M at 40 m), while ours is one frozen ray grid re-solved per range with spherical-wave phase. At short range on the minimum budget PathSolver costs less per pose, and cost crosses over as range or budget grows. Ledgers: report07_ray_budget_test.json, report07_sionna_ranges.json, report07_three_engine_ranges.json, report07_range40_raybudget.json, deck_ours_by_range.json.